Generative Modeling via Measure Transport

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Introduction

Working assumption: data = samples from an unknown target distribution $\rho_\star$.

Aim: learn this distribution in a *constructive way*, i.e. build a sampler of the target.

These notes show how to achieve this goal using *dynamic transportation of measure*. Specifically, we use the framework of **stochastic interpolants** to construct probability flow ODEs and SDEs that push a simple base distribution ρ_0 (e.g. a Gaussian) onto the target $\rho_\star$. Connections with **flow matching**, **rectified flows**, and **score-based diffusion** will be discussed throughout, along with generalizations to **multi-task interpolants**, **flow map matching**, and **generative models in discrete space** (with connections to transformers).

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1. Background and Setup

1.1. Probability Measures and Push-forwards

Push-forward. Consider the ODE

$$\dot{X}_t = b_t(X_t), \qquad X_{t=0} = x_0, \tag{1}$$

where $b : [0,1] \times \mathbb{R}^d \to \mathbb{R}^d$ is a *velocity field*. We assume b is measurable in t and, for each t, satisfies a global Lipschitz condition in x:

$$\sup_{t\in[0,1]} \|b_t\|_{\mathrm{Lip}} < \infty, \qquad \sup_{t\in[0,1]} \|b_t\|_{L^\infty} < \infty, \tag{2}$$

which guarantees that (1) has a unique global solution $X_t = X_t(x_0)$ for every initial condition $x_0 \in \mathbb{R}^d$ and all $t \in [0,1]$. These conditions will be relaxed later (see Remark 2) to accommodate velocity fields that blow up integrably near $t = 1$, as arises naturally when μ_1 concentrates on a low-dimensional set.

Given these conditions, for each t the map $x_0 \mapsto X_t(x_0)$ is a well-defined measurable map $\mathbb{R}^d \to \mathbb{R}^d$. The *push-forward* of μ_0 through X_t, denoted $\mu_t = (X_t)_\# \mu_0$, is defined as the unique measure satisfying

$$\int_{\mathbb{R}^d} \phi(x)\mu_t(dx) = \int_{\mathbb{R}^d} \phi\big(X_t(x_0)\big)\mu_0(dx_0), \qquad \forall \phi \in C_b^2(\mathbb{R}^d), \tag{3}$$

where $X_t(x_0)$ denotes the solution of (1) started at x_0. Equivalently, $(X_t)_\# \mu_0(A) = \mu_0(X_t^{-1}(A))$ for every Borel set $A \subseteq \mathbb{R}^d$.

1.2. The Continuity Equation

We now show that if $\mu_t = (X_t)_\# \mu_0$ and X_t satisfies the ODE (1), then μ_t solves the *continuity equation* (also called the *transport equation*)

$$\partial_t \mu_t + \nabla \cdot (b_t \mu_t) = 0, \qquad \mu_{t=0} = \mu_0, \tag{4}$$

in the weak sense.

Derivation. Fix $\phi \in C_b^2(\mathbb{R}^d)$ and define

$$F_t = \int_{\mathbb{R}^d} \phi(x)\mu_t(dx) = \int_{\mathbb{R}^d} \phi\big(X_t(x_0)\big)\mu_0(dx_0),$$

where we used the push-forward definition (3). Differentiating under the integral

sign (justified by the boundedness of $\nabla\phi$ and b_t) and applying the chain rule:

$$\begin{aligned}\frac{d}{dt}F_t &= \int_{\mathbb{R}^d} \nabla\phi\big(X_t(x_0)\big)\cdot \dot{X}_t(x_0)\mu_0(dx_0)\\ &= \int_{\mathbb{R}^d} \nabla\phi\big(X_t(x_0)\big)\cdot b_t\big(X_t(x_0)\big)\mu_0(dx_0),\end{aligned}$$

where in the last line we used (1). Changing variables back via the push-forward (3) with test function $x \mapsto \nabla\phi(x)\cdot b_t(x)$:

$$\frac{d}{dt}\int_{\mathbb{R}^d}\phi(x)\mu_t(dx) = \int_{\mathbb{R}^d}\nabla\phi(x)\cdot b_t(x)\mu_t(dx).$$

Integrating by parts (i.e. using $\int \nabla\phi\cdot b_t d\mu_t = -\int \phi\nabla\cdot(b_t d\mu_t)$ in the distributional sense), gives the weak form of (4):

$$\frac{d}{dt}\int_{\mathbb{R}^d}\phi(x)\mu_t(dx) = -\int_{\mathbb{R}^d}\phi(x)\nabla\cdot\big(b_t(x)\mu_t(dx)\big), \qquad \forall\phi\in C_b^2(\mathbb{R}^d). \tag{5}$$

The initial condition $\mu_{t=0} = \mu_0$ holds by definition of the push-forward at $t = 0$, since $X_0(x_0) = x_0$.

Remark 1 (Strong form of the continuity equation)

When μ_t has a density ρ_t, equation (4) takes the strong form

$$\partial_t\rho_t(x) + \nabla\cdot\big(b_t(x)\rho_t(x)\big) = 0,$$

which is the classical continuity equation from fluid mechanics.

Remark 2 (One-sided Lipschitz condition)

The uniform Lipschitz assumption (2) can be relaxed to a *one-sided Lipschitz* (OSL) condition:

$$\langle b_t(x) - b_t(y), x - y\rangle \le \lambda(t)|x-y|^2, \qquad \forall x, y\in\mathbb{R}^d, \tag{6}$$

for some $\lambda : [0,1) \to \mathbb{R}$ with $\int_0^1 \lambda^+(t)dt < \infty$, where $\lambda^+ = \max(\lambda, 0)$, together with a linear growth condition $|b_t(x)| \le C(t)(1+|x|)$ with $\int_0^1 C(t)dt < \infty$. Under these conditions, (1) still has a unique solution for all $t\in[0,1)$. This relaxation is important in practice: as we will see in Section 2, the velocity field $b_t(x) = \mathbb{E}[x_1 - x_0 \mid I_t = x]$ may blow up as $t\to 1$ when μ_1 concentrates on a low-dimensional set, but it does so in an integrable way that the OSL condition accommodates. The OSL condition controls only the *compressive* part of b_t: it permits trajectories to collapse (large negative $\lambda(t)$) while preventing exponential divergence.

1.3. Stochastic Differential Equations and the Fokker–Planck Equation

In later sections we will work not only with deterministic flows but also with stochastic dynamics. We collect here the key facts about stochastic differential equations (SDEs) that will be needed; proofs and detailed treatments can be found in [6, 7].

Itô SDEs. Let W_t be a standard d-dimensional Brownian motion on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$. An *Itô SDE* is an equation of the form

$$dX_t = b_t(X_t)dt + \sqrt{2\epsilon_t}dW_t, \qquad X_{t=0} \sim \mu_0, \tag{7}$$

where $b_t : \mathbb{R}^d \to \mathbb{R}^d$ is the *drift* and $\epsilon_t \geq 0$ is a (scalar) *diffusivity*. Under the same Lipschitz conditions as in (2), (7) has a unique strong solution.

Fokker–Planck equation. If X_t solves (7) and $\mu_t = \mathrm{Law}(X_t)$, then the density ρ_t of μ_t satisfies the *Fokker–Planck equation*

$$\partial_t \rho_t = -\nabla \cdot (b_t \rho_t) + \epsilon_t \Delta \rho_t, \qquad \rho_{t=0} = \rho_0. \tag{8}$$

This is the analogue of the continuity equation (4): setting $\epsilon_t = 0$ recovers (4) exactly. The Laplacian term $\epsilon_t \Delta \rho_t$ reflects the spreading of the density due to noise.

Derivation. It parallels that of (5): for $\phi \in C_b^2(\mathbb{R}^d)$, Itô's formula gives

$$d\phi(X_t) = \nabla\phi(X_t) \cdot b_t(X_t)dt + \epsilon_t \Delta\phi(X_t)dt + \sqrt{2\epsilon_t}\nabla\phi(X_t) \cdot dW_t.$$

Taking expectations kills the martingale term, yielding the weak form of (8):

$$\frac{d}{dt}\int \phi\rho_t dx = \int \big(\nabla\phi \cdot b_t + \epsilon_t \Delta\phi\big)\rho_t dx = \int \phi\big(-\nabla \cdot (b_t\rho_t) + \epsilon_t \Delta\rho_t\big)dx.$$

Backward Kolmogorov equation and Feynman–Kac formula. The Fokker–Planck equation (8) describes the evolution of the *marginal law* of X_t (a "forward" equation in the density). Its formal adjoint is the *backward Kolmogorov equation* (BKE), which governs how *observables* propagate backward in time. For a potential $V_t : \mathbb{R}^d \to \mathbb{R}$ and terminal condition $g : \mathbb{R}^d \to \mathbb{R}$, the BKE reads

$$\partial_t u_t + b_t \cdot \nabla u_t + \epsilon_t \, \Delta u_t - V_t \, u_t = 0, \qquad u_{t=1} = g. \tag{9}$$

Setting $V_t \equiv 0$ gives the pure BKE; adding $-V_t u_t$ is known as the BKE *with killing*. The *Feynman–Kac formula* identifies the solution of (9) as a conditional expectation along SDE paths.

Derivation. Define $M_s = u_s(X_s)\exp(-\int_t^s V_r(X_r)\,dr)$ for $s \in [t, 1]$, where X_s solves (7) with $X_t = x$. Applying Itô's formula to $u_s(X_s)$ and using (9) shows that the drift of M_s is

$$\big(\partial_s u_s + b_s \cdot \nabla u_s + \epsilon_s \, \Delta u_s - V_s \, u_s\big)(X_s)\, e^{-\int_t^s V_r \, dr} = 0,$$

so M_s is a local martingale (a true martingale under mild growth conditions on g and V). In particular $\mathbb{E}[M_1 \mid X_t = x] = M_t = u_t(x)$, which gives

$$u_t(x) \;=\; \mathbb{E}\left[g(X_1)\,\exp\left(-\int_t^1 V_s(X_s)\,ds\right)\,\middle|\, X_t = x\right]. \tag{10}$$

In the special case $V_t \equiv 0$, this reduces to $u_t(x) = \mathbb{E}[g(X_1) \mid X_t = x]$: the solution of the pure BKE is simply the conditional expectation of the terminal observable $g(X_1)$ given the current state $X_t = x$.

Girsanov theorem and path-space measures. Two SDEs with the same diffusion coefficient but different drifts b_t and $\hat{b}_t$ generate two path-space measures $\mathbb{P}$ and $\hat{\mathbb{P}}$ on $C([0,1];\mathbb{R}^d)$. The *Girsanov theorem* [6] gives an explicit formula for their Radon–Nikodým derivative. Under the Novikov condition $\mathbb{E}^{\hat{\mathbb{P}}}[\exp(\frac{1}{4}\int_0^1 \frac{|b_t-\hat{b}_t|^2}{\epsilon_t}\,dt)] < \infty$, we have $\mathbb{P} \ll \hat{\mathbb{P}}$ and

$$\begin{aligned}\frac{d\mathbb{P}}{d\hat{\mathbb{P}}}(X) &= \exp\left(\int_0^1 \frac{b_t(X_t)-\hat{b}_t(X_t)}{2\epsilon_t}\cdot\big(dX_t-\hat{b}_t(X_t)\,dt\big) - \frac{1}{4}\int_0^1 \frac{|b_t(X_t)-\hat{b}_t(X_t)|^2}{\epsilon_t}\,dt\right)\\ &= \exp\left(\int_0^1 \frac{b_t(X_t)-\hat{b}_t(X_t)}{\sqrt{2\epsilon_t}}\cdot dW_t^{\hat{\mathbb{P}}} - \frac{1}{4}\int_0^1 \frac{|b_t(X_t)-\hat{b}_t(X_t)|^2}{\epsilon_t}\,dt\right), \qquad (11)\end{aligned}$$

where X is a dummy path variable and $W_t^{\hat{\mathbb{P}}}$ is the Brownian motion driving the $\hat{\mathbb{P}}$-dynamics, related to X by $dW_t^{\hat{\mathbb{P}}} = \frac{1}{\sqrt{2\epsilon_t}}(dX_t - \hat{b}_t(X_t)\,dt)$. Taking the logarithm and expectation under $\mathbb{P}$ gives the *KL divergence between path-space measures*:

$$\mathrm{KL}(\mathbb{P}\,\|\,\hat{\mathbb{P}}) = \frac{1}{4}\int_0^1 \frac{1}{\epsilon_t}\,\mathbb{E}^{\mathbb{P}}\Big[|b_t(X_t)-\hat{b}_t(X_t)|^2\Big]\,dt. \qquad (12)$$

This formula will be central in Section 3, where we use it to measure the error introduced by replacing the true drift b_t with a learned approximation $\hat{b}_t$, and to derive the optimal diffusivity $\epsilon_t^\star$.

2. Generation via Probability Flow

2.1. Stochastic Interpolants: Definition and Basic Properties

Let (x_0, x_1) be a pair of random variables with $x_0 \sim \mu_0$ and $x_1 \sim \mu_1$. The joint distribution of (x_0, x_1) can be *any coupling* $\pi(dx_0, dx_1)$ of μ_0 and μ_1, i.e. any joint measure with the correct marginals:

$$\int \pi(dx_0, dx_1) = \mu_0(dx_0), \qquad \int \pi(dx_0, dx_1) = \mu_1(dx_1).$$

The simplest choice is the independent coupling $\pi = \mu_0 \otimes \mu_1$, but other couplings (e.g. the optimal transport map) may be used.

The *(linear) stochastic interpolant* is defined as

$$I_t = (1-t)\,x_0 + t\,x_1, \qquad t \in [0,1]. \qquad (13)$$

This is a random variable for each t, built from the pair (x_0, x_1). Its basic properties are immediate:

- **Boundary conditions:** $I_0 = x_0 \sim \mu_0$ and $I_1 = x_1 \sim \mu_1$, regardless of the coupling.
- **Linearity:** the path $t \mapsto I_t$ is a straight line in $\mathbb{R}^d$ connecting x_0 to x_1.
- **Time derivative:** $\dot{I}_t = x_1 - x_0$, which is constant along each trajectory.
- **Marginal law:** the law of I_t depends on the coupling π; different couplings yield different interpolating distributions $\mu_t = \mathrm{Law}(I_t)$.

The interpolating distribution $\mu_t = \mathrm{Law}(I_t)$ is characterized by

$$\int_{\mathbb{R}^d} \phi(x)\, \mu_t(dx) = \mathbb{E}\big[\phi(I_t)\big], \qquad \forall\, \phi \in C_b^2(\mathbb{R}^d), \tag{14}$$

where the expectation is taken over the joint law $\pi(dx_0, dx_1)$ of the pair (x_0, x_1).

2.2. The Continuity Equation for the Interpolant

We now show that $\mu_t = \mathrm{Law}(I_t)$ solves the continuity equation (4) with a specific velocity field determined by a conditional expectation.

Derivation. Fix $\phi \in C_b^2(\mathbb{R}^d)$. Differentiate $\mathbb{E}[\phi(I_t)]$ with respect to t, using the chain rule and $\dot{I}_t = x_1 - x_0$:

$$\frac{d}{dt}\, \mathbb{E}[\phi(I_t)] = \mathbb{E}\big[\nabla\phi(I_t) \cdot \dot{I}_t\big] = \mathbb{E}\big[\nabla\phi(I_t) \cdot (x_1 - x_0)\big].$$

We now apply the *tower property* of conditional expectation, conditioning on I_t:

$$\mathbb{E}\big[\nabla\phi(I_t) \cdot (x_1 - x_0)\big] = \mathbb{E}\Big[\nabla\phi(I_t) \cdot \mathbb{E}\big[x_1 - x_0 \,\big|\, I_t\big]\Big] = \mathbb{E}\big[\nabla\phi(I_t) \cdot b_t(I_t)\big],$$

where we define the *velocity field*

$$\boxed{b_t(x) = \mathbb{E}\big[x_1 - x_0 \,\big|\, I_t = x\big]. \qquad (15)}$$

Using the definition (14) to write the last expectation as an integral against μ_t:

$$\frac{d}{dt} \int_{\mathbb{R}^d} \phi(x)\, \mu_t(dx) = \int_{\mathbb{R}^d} \nabla\phi(x) \cdot b_t(x)\, \mu_t(dx).$$

This is exactly the weak form of the continuity equation (5), so μ_t satisfies

$$\partial_t \mu_t + \nabla \cdot (b_t\, \mu_t) = 0, \qquad \mu_{t=0} = \mu_0, \tag{16}$$

with velocity field b_t given by (15).

2.3. The Probability Flow ODE and Generation

Proposition (Probability flow ODE)

Let b_t be defined by (15) and assume it satisfies (2). Then any solution X_t of the ODE (1) with initial condition $X_{t=0} \sim \mu_0$ satisfies

$$\mathrm{Law}(X_t) = \mathrm{Law}(I_t) = \mu_t, \qquad \forall\, t \in [0,1]. \tag{17}$$

In particular, $\mathrm{Law}(X_1) = \mu_1$: the ODE pushes the base distribution exactly onto the target.

Proof. Both $\mu_t = \mathrm{Law}(I_t)$ and $\mathrm{Law}(X_t) = (X_t)_\# \mu_0$ satisfy the same continuity equation (16) with the same initial condition μ_0; uniqueness of solutions to the continuity equation (under (2)) then gives (17). □

We leave aside the question of the regularity of b_t defined by (15) for now, and return to it later.

Generation procedure. Equation (17) immediately suggests a generative algorithm:

1. **Train:** given samples from μ_1 (the data), approximate $b_t(x)$ by learning the conditional expectation (15), e.g. by minimizing the regression loss
$$\mathcal{L}(\hat{b}) = \int_0^1 \mathbb{E}\Big[\big|\hat{b}_t(I_t) - (x_1 - x_0)\big|^2\Big]\, dt.$$
2. **Sample:** draw $X_0 \sim \mu_0$ (the base, e.g. a standard Gaussian), then integrate the ODE $\dot{X}_t = b_t(X_t)$ from $t = 0$ to $t = 1$ using a numerical solver.
3. **Output:** $X_1 \approx x_1 \sim \mu_1$.

Remark 3 (General interpolants)

The linear interpolant (13) is a special case. More generally, one can take

$$I_t = \alpha_t\, x_0 + \beta_t\, x_1, \tag{18}$$

where $\alpha, \beta \in C^1([0,1])$ satisfy the boundary conditions

$$\alpha_0 = 1, \quad \alpha_1 = 0, \qquad \beta_0 = 0, \quad \beta_1 = 1,$$

so that $I_0 = x_0 \sim \mu_0$ and $I_1 = x_1 \sim \mu_1$ still hold. The time derivative is $\dot{I}_t = \dot{\alpha}_t\, x_0 + \dot{\beta}_t\, x_1$, and the velocity field becomes

$$b_t(x) = \mathbb{E}\big[\dot{I}_t \,\big|\, I_t = x\big] = \dot{\alpha}_t\, \mathbb{E}[x_0 \mid I_t = x] + \dot{\beta}_t\, \mathbb{E}[x_1 \mid I_t = x]. \tag{19}$$

This generality will be exploited later. Note in particular that α_t and β_t need not be scalar: they could be linear operators on $\mathbb{R}^d$ (e.g. time-dependent diagonal matrices or scalar multiples of the identity with different rates per coordinate), allowing anisotropic interpolation schedules.

Remark 4 (Relations between conditional expectations)

Define the three conditional fields

$$b_t(x) = \mathbb{E}[\dot{I}_t \mid I_t = x], \qquad \eta_t^0(x) = \mathbb{E}[x_0 \mid I_t = x], \qquad \eta_t^1(x) = \mathbb{E}[x_1 \mid I_t = x].$$

From (19) we have

$$b_t(x) = \dot{\alpha}_t\, \eta_t^0(x) + \dot{\beta}_t\, \eta_t^1(x). \tag{20}$$

Moreover, taking conditional expectations of $I_t = \alpha_t x_0 + \beta_t x_1$ given $I_t = x$:

$$x = \alpha_t\, \eta_t^0(x) + \beta_t\, \eta_t^1(x). \tag{21}$$

Equations (20)–(21) form a 2×2 linear system in (η_t^0, η_t^1). Provided $\alpha_t \dot{\beta}_t - \beta_t \dot{\alpha}_t \neq 0$ (which holds for the linear interpolant, where this quantity equals 1), any one of b_t, η_t^0, η_t^1 determines the other two. Solving, one finds for instance

$$\eta_t^1(x) = \frac{\dot{\alpha}_t\, x - \alpha_t\, b_t(x)}{\dot{\alpha}_t \beta_t - \alpha_t \dot{\beta}_t}, \qquad \eta_t^0(x) = \frac{x - \beta_t\, \eta_t^1(x)}{\alpha_t}.$$

In practice one may choose to regress any one of these three targets and recover the others algebraically.

Connection 1 — Flow matching

Flow matching [4, 1] constructs the same velocity field b_t by expressing the regression target through the *conditional* probability path $p_t(x \mid x_0, x_1)$ rather than the marginal μ_t. Concretely, it rewrites $b_t(x) = \mathbb{E}[\dot{I}_t \mid I_t = x]$ as an average of per-pair conditional velocity fields $u_t(x \mid x_0, x_1) = \dot{I}_t$ over the conditional law $p_t(\cdot \mid x_0, x_1)$. By the tower property of conditional expectation, this is identical to (15), and the two regression losses are equivalent up to an additive constant independent of the model. The resulting ODEs and generated distributions are therefore the same; the conditional formulation is simply a different but equivalent way to organize the training objective.

Connection 2 — Rectified flows

Rectified flows [5] are exactly the framework above with the linear interpolant (13) and the independent coupling $\pi = \mu_0 \otimes \mu_1$. The name refers to the observation that the ODE trajectories tend to be straighter than those of arbitrary couplings, reducing numerical integration error. The "rectification" step — iteratively resampling the coupling by pairing x_0 with the generated $X_1(x_0)$ and retraining — is an optional variance-reduction procedure that progressively straightens trajectories; its connection to optimal transport is discussed in Section **??**.

Connection 3 — Denoisers and DDPM

A natural alternative to regressing b_t directly is to regress the *denoiser* $\eta_t^1(x) = \mathbb{E}[x_1 \mid I_t = x]$, which predicts the clean data point x_1 from the noisy interpolant $I_t = x$. By the relations of Remark 4, η_t^1 and b_t determine each other algebraically, so both are

equivalent training targets.
More generally, one can consider the conditional expectation of a future interpolant value given the present one. For $0 \le t < s \le 1$, writing $I_s = \alpha_s x_0 + \beta_s x_1$ and taking the conditional expectation given $I_t = x$:

$$\mathbb{E}[I_s \mid I_t = x] \;=\; \alpha_s\, \eta_t^0(x) + \beta_s\, \eta_t^1(x), \tag{22}$$

which is again a linear combination of η_t^0 and η_t^1, fully determined by b_t via (20)–(21). This suggests an iterative generation scheme on a grid $0 = t_0 < t_1 < \cdots < t_N = 1$: starting from $X_0 \sim \mu_0$, set

$$X_{t_{k+1}} \;=\; \mathbb{E}[I_{t_{k+1}} \mid I_{t_k} = X_{t_k}], \qquad k = 0, 1, \dots, N-1.$$

This is precisely the **DDPM** ancestral sampling scheme [3]. Two observations are key:

- **At finite step size** $h = t_{k+1} - t_k > 0$, this is *not* an exact generative model: the iterated conditional means do not produce samples from μ_1, because $\mathbb{E}[I_s \mid I_t]$ is a deterministic map and collapses the randomness in x_1.
- **In the limit** $h \to 0$, the scheme becomes exact. Indeed, by (22),

$$\lim_{h\to 0} \frac{\mathbb{E}[I_{t+h} \mid I_t = x] - x}{h} \;=\; \dot\alpha_t\, \eta_t^0(x) + \dot\beta_t\, \eta_t^1(x) \;=\; b_t(x),$$

  so the incremental denoiser converges to the velocity field b_t, and the iterative scheme converges to the probability flow ODE (1).

DDPM can therefore be understood as a discretization of the probability flow ODE, with the step-size bias being the price paid for using a finite grid.

Connection 4 — Score-Based Diffusion Models via the OU Process

A classical construction of a probability path begins with the *Ornstein–Uhlenbeck* (OU) process in continuous time $\tau \ge 0$:

$$d\tilde X_\tau = -\tilde X_\tau\, d\tau + \sqrt{2}\, dW_\tau, \qquad \tilde X_0 = x_1, \tag{23}$$

started at a data point $x_1 \sim \mu_1$. Solving (23) conditionally on x_1 gives $\tilde X_\tau | x_1 \sim \mathcal{N}(e^{-\tau} x_1, (1 - e^{-2\tau}) I_d)$, so the mean decays toward zero and the variance grows toward 1 as $\tau \to \infty$. Reparametrizing by $t = e^{-\tau}$ (so $t = 1$ corresponds to $\tau = 0$, i.e. the data, and $t = 0$ corresponds to $\tau \to \infty$, i.e. pure noise) the conditional distribution becomes

$$\tilde X_\tau | x_1 \sim \mathcal{N}(t\, x_1, (1 - t^2) I_d).$$

This is exactly the stochastic interpolant $I_t = \alpha_t x_0 + \beta_t x_1$ with

$$\alpha_t = \sqrt{1 - t^2}, \qquad \beta_t = t, \tag{24}$$

and $x_0 \sim \mathcal{N}(0, I_d)$ independent of x_1, which satisfies $\alpha_t^2 + \beta_t^2 = 1$. The VP-SDE of [8] corresponds to a closely related time change of the same OU process.

Once (α_t, β_t) are identified, the velocity field b_t and the probability flow ODE (1) follow immediately: no further reference to the OU process is needed for generation. The SDE counterpart, which adds noise back into the generative process, will be taken up in Section 3.

This rederivation highlights a key conceptual gain of the stochastic interpolant framework. In score-based diffusion models [8, 3], the *probability path* (determined by the forward noising SDE) and the *inference procedure* (the reverse SDE or probability flow ODE) are tightly coupled: changing the forward process requires re-deriving and retraining the reverse. In the SI framework these two choices are *decoupled*: the interpolant (α_t, β_t) defines the probability path and determines the regression target for b_t, while the choice of how to sample from this path at inference time — deterministically via the probability flow ODE (1), or stochastically via a family of SDEs parametrized by a diffusivity ϵ_t — is a free parameter that can be tuned independently without any retraining (see Section 3). The OU-derived interpolant (24) is simply one convenient choice among many; the framework does not privilege it.

2.4. Density Evolution and Log-Likelihood

We now suppose that μ_0 and μ_1 both have strictly positive densities ρ_0 and ρ_1 with respect to the Lebesgue measure. Under mild regularity conditions on the coupling π and the velocity field b_t, the interpolant measure $\mu_t = \mathrm{Law}(I_t)$ also has a positive density ρ_t for all $t \in (0, 1)$. Using physicist's notation for the Dirac delta $\delta(\cdot)$, it can be written formally as

$$\rho_t(x) = \int_{\mathbb{R}^d \times \mathbb{R}^d} \delta\big(x - (1-t)x_0 - tx_1\big)\, \pi(dx_0, dx_1),$$

which is shorthand for the proper definition via test functions: for all $\phi \in C_b^2(\mathbb{R}^d)$,

$$\int_{\mathbb{R}^d} \phi(x)\, \rho_t(x)\, dx = \int_{\mathbb{R}^d \times \mathbb{R}^d} \phi\big((1-t)x_0 + tx_1\big)\, \pi(dx_0, dx_1) = \mathbb{E}\big[\phi(I_t)\big],$$

which is simply the definition (14) of μ_t.

Log-density along the flow. Let X_t solve the probability flow ODE (1) with velocity b_t. Since b_t is Lipschitz, the flow map $x_0 \mapsto X_t(x_0)$ is a diffeomorphism for each t, and the density ρ_t evolves according to the strong continuity equation

$$\partial_t \rho_t + \nabla \cdot (b_t\, \rho_t) = 0.$$

Expanding the divergence: $\partial_t \rho_t + b_t \cdot \nabla \rho_t + \rho_t\, \nabla \cdot b_t = 0$. Dividing by $\rho_t > 0$:

$$\partial_t \log \rho_t + b_t \cdot \nabla \log \rho_t + \nabla \cdot b_t = 0,$$

which can be written as a total derivative along trajectories of the ODE,

$$\frac{d}{dt} \log \rho_t(X_t) = -\nabla \cdot b_t(X_t).$$

Integrating from s to t yields the following result.

Proposition (Log-density transport)

Let μ_0, μ_1 have strictly positive densities ρ_0, ρ_1, and suppose $\mu_t = \mathrm{Law}(I_t)$ has a positive density ρ_t for $t \in (0,1)$. Let X_t solve (1) with Lipschitz velocity b_t. Then, for all $0 \le s \le t \le 1$,

$$\log \rho_t(X_t) = \log \rho_s(X_s) - \int_s^t \nabla \cdot b_\tau(X_\tau)\, d\tau. \tag{25}$$

Log-likelihood estimation. Equation (25) is useful whenever $\log \rho_0$ is known in closed form — the canonical example being a standard Gaussian base, $\rho_0(x) = \mathcal{N}(x;\, 0, I_d)$. It can be exploited in two complementary directions.

Log-likelihood of data points. Given a data point $x_1 \sim \mu_1$, let $X_t(x_1)$ denote the solution of the ODE (1) with *final* condition $X_{t=1} = x_1$, integrated backward in time. Applying (25) with $s = 0$, $t = 1$:

$$\log \rho_1(x_1) = \log \rho_0\big(X_0(x_1)\big) - \int_0^1 \nabla \cdot b_t\big(X_t(x_1)\big)\, dt,$$

where $X_0(x_1) \in \mathbb{R}^d$ is the base-space point reached by the backward flow. Since ρ_0 is known, this gives the exact log-likelihood of any data point x_1.

Log-likelihood of generated points. Given a base sample $x_0 \sim \mu_0$, let $X_t(x_0)$ denote the solution of the ODE with *initial* condition $X_{t=0} = x_0$, integrated forward in time. Applying (25) with $s = 0$, $t = 1$:

$$\log \rho_1\big(X_1(x_0)\big) = \log \rho_0(x_0) - \int_0^1 \nabla \cdot b_t\big(X_t(x_0)\big)\, dt,$$

where $X_1(x_0) \in \mathbb{R}^d$ is the generated sample. This allows one to attach a likelihood score to each generated point.

In both cases, the divergence $\nabla \cdot b_t(X_t)$ can be estimated efficiently in high dimensions via Hutchinson's trace estimator, giving exact log-likelihoods up to numerical ODE integration error.

Remark 5 (Connection to normalizing flows)

The formula (25) with $s = 0$, $t = 1$ is equivalent to the classical *change-of-variables formula* for normalizing flows. Along the forward flow $X_t(x_0)$, the accumulated divergence equals the log-Jacobian of the flow map:

$$\int_0^1 \nabla \cdot b_t\big(X_t(x_0)\big)\, dt = \log \left|\det \nabla_{x_0} X_1(x_0)\right|,$$

so that

$$\log \rho_1\big(X_1(x_0)\big) = \log \rho_0(x_0) - \log \left|\det \nabla_{x_0} X_1(x_0)\right|,$$

which is precisely the change-of-variables formula, with the continuous-time ODE providing a tractable alternative to computing the Jacobian determinant directly.

2.5. Transport cost bound

The probability flow ODE does not in general produce an optimal transport map, but the total kinetic energy of the flow admits a clean upper bound in terms of the interpolant.

Proposition (Transport cost)

Let $X_t(x_0)$ solve $\dot{X}_t = b_t(X_t)$ with $X_{t=0} = x_0 \sim \mu_0$. Then

$$\mathbb{E}\left[|X_1(x_0) - x_0|^2\right] \;\leq\; \int_0^1 \mathbb{E}\left[|\dot{I}_t|^2\right] dt = \int_0^1 \mathbb{E}\left[|\dot{\alpha}_t x_0 + \dot{\beta}_t x_1|^2\right] dt. \tag{26}$$

Proof. Using $X_1(x_0) - x_0 = \int_0^1 \dot{X}_t(x_0)\, dt$ and the Cauchy–Schwarz inequality:

$$|X_1(x_0) - x_0|^2 = \left|\int_0^1 \dot{X}_t(x_0)\, dt\right|^2 \;\leq\; \int_0^1 |\dot{X}_t(x_0)|^2\, dt = \int_0^1 |b_t(X_t(x_0))|^2\, dt.$$

Taking expectations and using $\mathrm{Law}(X_t(x_0)) = \mathrm{Law}(I_t) = \mu_t$:

$$\mathbb{E}\left[|X_1(x_0) - x_0|^2\right] \;\leq\; \int_0^1 \mathbb{E}\left[|b_t(I_t)|^2\right] dt = \int_0^1 \mathbb{E}\left[|\mathbb{E}[\dot{I}_t \mid I_t]|^2\right] dt.$$

By Jensen's inequality applied to the conditional expectation:

$$\mathbb{E}\left[|\mathbb{E}[\dot{I}_t \mid I_t]|^2\right] \;\leq\; \mathbb{E}\left[\mathbb{E}[|\dot{I}_t|^2 \mid I_t]\right] = \mathbb{E}\left[|\dot{I}_t|^2\right].$$

Combining gives (26). □

2.6. Estimation Error and Stability

In practice the velocity field b_t is replaced by a learned approximation $\hat{b}_t$. Let $\hat{X}_t$ solve the ODE $\dot{\hat{X}}_t = \hat{b}_t(\hat{X}_t)$ with $\hat{X}_{t=0} = X_{t=0} \sim \mu_0$, and write $\hat{\mu}_t = \mathrm{Law}(\hat{X}_t)$. We bound the discrepancy between μ_t and $\hat{\mu}_t$ in the 2-Wasserstein distance

$$W_2(\mu_t, \hat{\mu}_t)^2 = \inf_{\gamma} \int |x - y|^2\, \gamma(dx, dy),$$

where the infimum is over couplings γ of $(\mu_t, \hat{\mu}_t)$.

Proposition (Stability of the flow)

Assume $\hat{b}_t$ is globally Lipschitz in x, uniformly in t:

$$\sup_{t\in[0,1]} \|\hat{b}_t\|_{\mathrm{Lip}} \;\le\; L.$$

Then, for all $t \in [0,1]$,

$$W_2^2(\mu_t, \hat{\mu}_t) \;\le\; 2e^{2Lt} \int_0^t \mathbb{E}\big[|b_s(X_s) - \hat{b}_s(X_s)|^2\big]\, ds. \tag{27}$$

Proof. Since $X_0 = \hat{X}_0$ (same initial condition), $W_2(\mu_t, \hat{\mu}_t)^2 \le \mathbb{E}[|X_t - \hat{X}_t|^2]$. Write $e_t = X_t - \hat{X}_t$; then

$$\frac{d}{dt} e_t = b_t(X_t) - \hat{b}_t(\hat{X}_t) = \big[b_t(X_t) - \hat{b}_t(X_t)\big] + \big[\hat{b}_t(X_t) - \hat{b}_t(\hat{X}_t)\big].$$

By the Lipschitz condition on $\hat{b}_t$ and Young's inequality ($2ab \le a^2 + b^2$):

$$\frac{d}{dt}|e_t|^2 \;\le\; 2L\,|e_t|^2 + 2\,|b_t(X_t) - \hat{b}_t(X_t)|^2.$$

Taking expectations and applying Grönwall's inequality with $e_0 = 0$:

$$\mathbb{E}[|e_t|^2] \;\le\; 2e^{2Lt} \int_0^t \mathbb{E}\big[|b_s(X_s) - \hat{b}_s(X_s)|^2\big]\, ds,$$

which implies (27). □

Interpretation. The bound (27) says that the W_2 error in the generated distribution grows at most exponentially in the Lipschitz constant L of $\hat{b}_t$, and linearly (before the exponential) in the L^2 approximation error $\int_0^1 \mathbb{E}[|b_t - \hat{b}_t|^2]\, dt$. In particular, if $\hat{b}_t \to b_t$ in L^2, then $\hat{\mu}_t \to \mu_t$ in W_2, and the quality of the generated samples is controlled directly by the regression loss used to train $\hat{b}_t$.

Concretely, recall that $b_t(x) = \mathbb{E}[\dot{I}_t \mid I_t = x]$ is the conditional expectation of $\dot{I}_t$ given $I_t = x$, so the regression loss satisfies

$$\begin{aligned}\mathcal{L}(\hat{b}) &= \int_0^1 \mathbb{E}\big[|\hat{b}_t(I_t) - \dot{I}_t|^2\big]\, dt \\ &= \int_0^1 \mathbb{E}\big[|\hat{b}_t(I_t) - b_t(I_t)|^2\big]\, dt + \int_0^1 \Big(\mathbb{E}\big[|\dot{I}_t|^2\big] - \mathbb{E}\big[|b_t(I_t)|^2\big]\Big)\, dt,\end{aligned}$$

where the second term is a constant (independent of $\hat{b}$) equal to the variance of $\dot{I}_t$ given I_t, integrated over time. Minimizing $\mathcal{L}(\hat{b})$ over $\hat{b}$ is therefore equivalent to minimizing $\int_0^1 \mathbb{E}[|\hat{b}_t(I_t) - b_t(I_t)|^2]\, dt$, which is precisely the L^2 approximation error appearing in (27). The additive constant $\int_0^1 (\mathbb{E}[|\dot{I}_t|^2] - \mathbb{E}[|b_t(I_t)|^2])\, dt \ge 0$ is unknown but does not affect the minimizer; it can be interpreted as the irreducible variance of the interpolant velocity that no deterministic velocity field can capture.

3. Generation via Diffusions

The probability flow ODE of Section 2 is deterministic: given the initial condition $X_0 \sim \mu_0$, the trajectory is fully determined by the velocity field b_t. This section shows that infinitely many *stochastic* processes — Itô diffusions — share the same marginal laws μ_t, and that the degree of stochasticity can be tuned freely via a diffusivity $\epsilon_t > 0$. The key ingredient is a formula for the *score* $\nabla \log \rho_t$ in terms of the velocity field b_t, which holds whenever the base distribution μ_0 is Gaussian.

3.1. The Score via Tweedie's Formula

Assume throughout this section that the base distribution is a standard Gaussian:

$$\mu_0 = \mathcal{N}(0, I_d), \qquad \rho_0(x) = (2\pi)^{-d/2} e^{-|x|^2/2}. \tag{28}$$

We work with the general interpolant $I_t = \alpha_t x_0 + \beta_t x_1$ of (18). Since $x_0 \sim \mathcal{N}(0, I_d)$ and $x_1 \sim \mu_1$ are independent under the coupling $\pi = \mu_0 \otimes \mu_1$, the conditional distribution of I_t given x_1 is Gaussian: $I_t \mid x_1 \sim \mathcal{N}(\beta_t x_1,\ \alpha_t^2 I_d)$. The marginal density is therefore the mixture

$$\rho_t(x) = \int \mathcal{N}(x;\ \beta_t x_1,\ \alpha_t^2 I_d)\, \mu_1(dx_1).$$

Tweedie's formula. Differentiating the mixture representation with respect to x:

$$\nabla_x \rho_t(x) = -\frac{1}{\alpha_t^2} \int (x - \beta_t x_1)\, \mathcal{N}(x;\ \beta_t x_1,\ \alpha_t^2 I_d)\, \mu_1(dx_1).$$

Dividing by $\rho_t(x)$ and recognizing the conditional expectation $\mathbb{E}[x - \beta_t x_1 \mid I_t = x] = x - \beta_t\, \eta_t^1(x)$:

$$\nabla_x \log \rho_t(x) = -\frac{x - \beta_t\, \eta_t^1(x)}{\alpha_t^2}.$$

Using (21), namely $x = \alpha_t\, \eta_t^0(x) + \beta_t\, \eta_t^1(x)$, the numerator simplifies to $\alpha_t\, \eta_t^0(x)$:

Proposition (Score via Tweedie's formula)

Assume $\mu_0 = \mathcal{N}(0, I_d)$ and $I_t = \alpha_t x_0 + \beta_t x_1$ with $\alpha_t > 0$ for $t \in [0, 1)$. Then the score of the interpolant density ρ_t satisfies

$$\nabla \log \rho_t(x) = -\frac{\eta_t^0(x)}{\alpha_t} = -\frac{\mathbb{E}[x_0 \mid I_t = x]}{\alpha_t}. \tag{29}$$

Since η_t^0 is expressible in terms of b_t via the linear system (20)–(21), the score (29) is fully determined by the velocity field alone. We introduce the shorthand

$$\gamma_t := \alpha_t \dot{\beta}_t - \dot{\alpha}_t \beta_t$$

for the Wronskian of (α_t, β_t). Solving the system (20)–(21) for η_t^0 explicitly:

$$\eta_t^0(x) = \frac{\dot{\beta}_t\, x - \beta_t\, b_t(x)}{\gamma_t}, \tag{30}$$

and substituting into (29):

$$\nabla \log \rho_t(x) = \frac{\beta_t\, b_t(x) - \dot{\beta}_t\, x}{\alpha_t\, \gamma_t}. \tag{31}$$

In the linear case $\alpha_t = 1 - t$, $\beta_t = t$, we have $\gamma_t = 1$, and (31) simplifies to

$$\nabla \log \rho_t(x) = \frac{t\, b_t(x) - x}{1 - t}, \qquad t \in [0, 1).$$

Remark 6 (Apparent singularity at $t = 1$)

The score (29) involves $1/\alpha_t$, which diverges as $\alpha_t \to 0$ (i.e. as $t \to 1$). This is not a pathology of the formula but a fundamental limitation: Tweedie's formula derives the score by exploiting the Gaussian noise $\alpha_t x_0$ injected into the interpolant $I_t = \alpha_t x_0 + \beta_t x_1$. At $t = 1$ this noise has vanished entirely ($\alpha_1 = 0$), so the formula loses all information about x_0 and the estimator for the score becomes undefined (even if the score itself is not). In the SDE formulation below, the diffusivity ϵ_t must therefore vanish at least as fast as α_t to prevent the score-corrected drift from diverging.

3.2. From the Continuity Equation to the Fokker–Planck Equation

The continuity equation (16) can be rewritten as a *Fokker–Planck equation* by adding and subtracting a diffusion term. Let $\epsilon_t > 0$ be any smooth function of t, playing the role of a time-dependent diffusivity. Using the identity $\epsilon_t\, \Delta\rho_t = \epsilon_t\, \nabla \cdot (\rho_t\, \nabla \log \rho_t)$, we add and subtract this term from (16):

$$\begin{aligned}
\partial_t \rho_t &= -\nabla \cdot (b_t\, \rho_t) \\
&= -\nabla \cdot (b_t\, \rho_t) - \epsilon_t\, \nabla \cdot (\rho_t\, \nabla \log \rho_t) + \epsilon_t\, \Delta\rho_t \\
&= -\nabla \cdot \big((b_t - \epsilon_t\, \nabla \log \rho_t)\, \rho_t\big) + \epsilon_t\, \Delta\rho_t.
\end{aligned}$$

This is a Fokker–Planck equation

$$\partial_t \rho_t + \nabla \cdot \big(b_t^\epsilon\, \rho_t\big) = \epsilon_t\, \Delta\rho_t, \tag{32}$$

with diffusion coefficient $\sqrt{2\epsilon_t}$ and *score-corrected drift*

$$b_t^\epsilon(x) = b_t(x) - \epsilon_t\, \nabla \log \rho_t(x). \tag{33}$$

Substituting (31) to eliminate the score in favour of b_t:

$$b_t^\epsilon(x) = \left(1 + \frac{\epsilon_t \beta_t}{\alpha_t \gamma_t}\right) b_t(x) - \frac{\epsilon_t \dot{\beta}_t}{\alpha_t \gamma_t} x. \tag{34}$$

For the linear interpolant ($\alpha_t = 1 - t$, $\beta_t = t$, $\gamma_t = 1$) this becomes

$$b_t^\epsilon(x) = \left(1 + \frac{\epsilon_t t}{1 - t}\right) b_t(x) - \frac{\epsilon_t}{1 - t} x, \qquad t \in [0, 1).$$

3.3. The SDE Family and Generation

The Fokker–Planck equation (32) is the forward Kolmogorov equation of the Itô SDE

$$dX_t = b_t^\epsilon(X_t)\, dt + \sqrt{2\epsilon_t}\, dW_t, \qquad X_0 \sim \mu_0, \tag{35}$$

where W_t is a standard d-dimensional Brownian motion. By standard results in stochastic analysis, the law of X_t solving (35) satisfies (32), and hence $\mathrm{Law}(X_t) = \mu_t$ for all $t \in [0, 1]$. This gives an entire *family* of generative diffusion processes, one for each choice of $\epsilon_t > 0$, all sharing the same marginal laws.

Regularity of b_t^ϵ near $t = 1$. From (34), the score-corrected drift involves ϵ_t/α_t, which diverges as $t \to 1$ unless ϵ_t decays fast enough. For b_t^ϵ to remain bounded, we need $\epsilon_t = O(\alpha_t)$ as $t \to 1$; the simplest family satisfying this is

$$\epsilon_t = c\, \alpha_t, \qquad c > 0, \tag{36}$$

for which $\epsilon_t/\alpha_t = c$. Under this choice, (34) remains well-behaved all the way to $t = 1$.

Remark 7 (Euler–Maruyama discretization of the SDE)

The SDE (35) is integrated numerically by the Euler–Maruyama scheme on a grid $0 = t_0 < t_1 < \cdots < t_N = 1$ with step $h = t_{k+1} - t_k$:

$$X_{t_{k+1}} = X_{t_k} + b_{t_k}^\epsilon(X_{t_k})\, h + \sqrt{2\epsilon_{t_k} h}\, Z_k,$$

with $Z_k \overset{\text{iid}}{\sim} \mathcal{N}(0, I_d)$ and $X_{t_0} \sim \mu_0$. The scheme has $O(h)$ weak error and $O(\sqrt{h})$ strong error. In the deterministic limit $\epsilon_t \equiv 0$ the noise term vanishes and one recovers the forward Euler discretization of the probability flow ODE (1), which underlies the DDPM connection of Connection 3: the iterative denoiser $\mathbb{E}[I_{t_{k+1}} \mid I_{t_k}]$ is precisely the mean of the Euler–Maruyama step with $\epsilon_t \equiv 0$, and the two coincide in the limit $h \to 0$.

3.4. Optimal Diffusivity via Girsanov's Theorem

In practice, the drift b_t^ϵ is replaced by a learned approximation $\hat{b}_t^\epsilon$, obtained by plugging a learned $\hat{b}_t \approx b_t$ into (34). We use *Girsanov's theorem* to quantify the resulting error and then optimize ϵ_t to minimize it.

Path measure KL via Girsanov. Let $\mathbb{P}^\epsilon$ be the path-space law of the exact SDE (35) (drift b_t^ϵ), and $\hat{\mathbb{P}}^\epsilon$ the law of the approximate SDE with drift $\hat{b}_t^\epsilon$ and the *same* diffusivity ϵ_t. By Girsanov's theorem:

Proposition (Path KL via Girsanov)

$$\mathrm{KL}\big(\mathbb{P}^\epsilon \,\|\, \hat{\mathbb{P}}^\epsilon\big) = \int_0^1 \frac{1}{4\epsilon_t} \mathbb{E}_{\mathbb{P}^\epsilon}\left[\left|b_t^\epsilon(X_t) - \hat{b}_t^\epsilon(X_t)\right|^2\right] dt.$$

Since $\mathrm{KL}(\mathrm{Law}(X_1) \,\|\, \mathrm{Law}(\hat{X}_1)) \le \mathrm{KL}(\mathbb{P}^\epsilon \,\|\, \hat{\mathbb{P}}^\epsilon)$ by the data-processing inequality, minimizing the path KL controls the KL between generated and target distributions at $t = 1$.

Optimizing ϵ_t. Since both b_t^ϵ and $\hat{b}_t^\epsilon$ are obtained from (34) by using b_t and $\hat{b}_t$ respectively, their difference is

$$b_t^\epsilon(x) - \hat{b}_t^\epsilon(x) = \left(1 + \frac{\epsilon_t \beta_t}{\alpha_t \gamma_t}\right)\big(b_t(x) - \hat{b}_t(x)\big).$$

Defining the pointwise squared approximation error $\delta_t^2 := \mathbb{E}[|b_t(X_t) - \hat{b}_t(X_t)|^2]$ and $A_t := \beta_t/(\alpha_t \gamma_t)$, the path KL becomes

$$\mathrm{KL}\big(\mathbb{P}^\epsilon \,\|\, \hat{\mathbb{P}}^\epsilon\big) \asymp \frac{1}{4}\int_0^1 \frac{(1 + A_t\,\epsilon_t)^2}{\epsilon_t}\,\delta_t^2\,dt.$$

Minimizing the integrand pointwise over $\epsilon_t > 0$, we write $f(\epsilon) = (1 + A_t\epsilon)^2/\epsilon = 1/\epsilon + 2A_t + A_t^2\epsilon$. Then $f'(\epsilon) = -1/\epsilon^2 + A_t^2 = 0$ gives:

Proposition (Optimal diffusivity)

The diffusivity minimizing the path KL pointwise in t is

$$\epsilon_t^\star = \frac{1}{|A_t|} = \frac{|\alpha_t\,\gamma_t|}{|\beta_t|} = \frac{\alpha_t\,|\alpha_t\dot{\beta}_t - \dot{\alpha}_t\beta_t|}{|\beta_t|}. \tag{37}$$

This depends only on the interpolation schedule (α_t, β_t) and not on the target distribution μ_1.

Linear interpolant. For $\alpha_t = 1 - t$, $\beta_t = t$, $\gamma_t = 1$:

$$\epsilon_t^\star = \frac{1-t}{t},$$

so the optimal noise amplitude $\sqrt{2\epsilon_t^\star} = \sqrt{2(1-t)/t}$ is large near $t = 0$ (where the base distribution is diffuse) and vanishes as $t \to 1$ (where the distribution concentrates near the target), as intuitively expected.

Interpretation. The optimal $\epsilon_t^\star$ balances two competing effects. Taking $\epsilon_t \to 0$ recovers the ODE ($b_t^\epsilon \to b_t$), and the path KL diverges like $1/\epsilon_t$ for any fixed drift error.

Taking $\epsilon_t \to \infty$ amplifies the drift error through the score-correction term. The minimum at $\epsilon_t^\star = 1/|A_t|$ is the unique balance, depending only on (α_t, β_t) and not on the unknown target μ_1.

3.5. Reverse-Time SDEs

Having established that the SDE (35) has marginal law $\mathrm{Law}(X_t) = \mu_t$, we now ask: is there a process whose law at time t equals μ_{1-t}, i.e. that runs the probability path backward? There are two natural ways to make this precise, which we distinguish carefully.

Forward-time reversal. Define $\rho_t^R = \rho_{1-t}$, i.e. the time-reversed density. Substituting $t \mapsto 1-t$ in the Fokker–Planck equation (32) gives

$$\partial_t \rho_t^R = \nabla \cdot (b_{1-t}^\epsilon \, \rho_t^R) - \epsilon_{1-t} \, \Delta \rho_t^R.$$

The right-hand side has the wrong sign on the diffusion term, so this is not a valid Fokker–Planck equation. To remedy this, we add and subtract $\epsilon_{1-t} \Delta \rho_t^R$ and use $\Delta \rho_t^R = \nabla \cdot (\rho_t^R \, \nabla \log \rho_t^R)$ to rewrite:

$$\begin{aligned} \partial_t \rho_t^R &= \nabla \cdot (b_{1-t}^\epsilon \, \rho_t^R) - 2\epsilon_{1-t} \, \nabla \cdot (\rho_t^R \, \nabla \log \rho_t^R) + \epsilon_{1-t} \, \Delta \rho_t^R \\ &= -\nabla \cdot \Big(\big(-b_{1-t}^\epsilon(x) + 2\epsilon_{1-t} \, \nabla \log \rho_t^R(x) \big) \rho_t^R \Big) + \epsilon_{1-t} \, \Delta \rho_t^R. \end{aligned}$$

This is a valid Fokker–Planck equation for a diffusion process X_t^R run *forward* from $t = 0$ to $t = 1$:

$$dX_t^R = \big(-b_{1-t}^\epsilon(X_t^R) + 2\epsilon_{1-t} \, \nabla \log \rho_{1-t}(X_t^R) \big) dt + \sqrt{2\epsilon_{1-t}} \, dW_t, \qquad X_0^R \sim \mu_1, \tag{38}$$

with the property that $\mathrm{Law}(X_t^R) = \mu_{1-t}$ for all $t \in [0,1]$, so in particular $X_1^R \sim \mu_0$. The process X^R is adapted to the forward filtration generated by W_t and can be simulated by a standard forward-time SDE solver, at the cost of evaluating the score $\nabla \log \rho_{1-t}$ at the time-reversed argument $1-t$.

Reverse-time SDE in the original time variable. Alternatively, one may keep the original time variable $t \in [0,1]$ and run the process backward from $t = 1$ to $t = 0$. By Anderson's time-reversal theorem [2], the process X_{1-t}^R satisfies the SDE

$$dX_t = \big(b_t^\epsilon(X_t) - 2\epsilon_t \, \nabla \log \rho_t(X_t) \big) dt + \sqrt{2\epsilon_t} \, d\bar{W}_t, \qquad X_1 \sim \mu_1, \tag{39}$$

run from $t = 1$ to $t = 0$, where $\bar{W}_t$ is a reverse-time Brownian motion adapted to the reverse filtration $\bar{\mathcal{F}}_t = \sigma(X_s : s \geq t)$. The solution satisfies $\mathrm{Law}(X_t) = \mu_t$ for all t, the same marginals as the forward SDE (35), and in particular $X_0 \sim \mu_0$. This form

is mathematically equivalent to (38) and is equally straightforward to simulate, as explained in Remark 8 below. Since the score $\nabla \log \rho_t$ is expressible in terms of b_t via (31), both (38) and (39) are fully parametrized by the single learned field $\hat{b}_t$.

Remark 8 (Euler–Maruyama discretization of the time-reversed SDE)

Both (38) and (39) are integrated numerically by standard Euler–Maruyama schemes on a grid $0 = t_0 < t_1 < \cdots < t_N = 1$ with step $h = t_{k+1} - t_k$. For the forward process (38), one steps from $t = 0$ to $t = 1$ in the usual way:

$$X^R_{t_{k+1}} = X^R_{t_k} + \big(-b^{\epsilon}_{1-t_k}(X^R_{t_k}) + 2\epsilon_{1-t_k} \nabla \log \rho_{1-t_k}(X^R_{t_k})\big)h + \sqrt{2\epsilon_{1-t_k}\, h}\, Z_k,$$

with $Z_k \overset{\text{iid}}{\sim} \mathcal{N}(0, I_d)$. For the reverse-time process (39), one steps backward from $t = 1$ to $t = 0$, replacing h by $-h$:

$$X_{t_k} = X_{t_{k+1}} - \big(b^{\epsilon}_{t_{k+1}}(X_{t_{k+1}}) - 2\epsilon_{t_{k+1}} \nabla \log \rho_{t_{k+1}}(X_{t_{k+1}})\big)h + \sqrt{2\epsilon_{t_{k+1}}\, h}\, Z_k.$$

Both schemes have the same cost per step and the same $O(h)$ weak error; the only difference is the direction of traversal and the sign of the drift increment. In practice one stores the grid in reverse order and applies the same update rule, so no special treatment of the reverse-time filtration is needed at the implementation level.

Connection 5 — Score-Based Diffusion Models

Score-based diffusion models [3, 8] are naturally recovered within the SI framework by starting from the OU process in continuous time $\tau \geq 0$ (Connection 4):

$$d\tilde{X}_\tau = -\tilde{X}_\tau\, d\tau + \sqrt{2}\, dW_\tau, \qquad \tilde{X}_0 = x_1 \sim \mu_1,$$

whose marginal is $\mathcal{N}(e^{-\tau}x_1, (1 - e^{-2\tau})I_d)$ conditionally on x_1. Time-reversing via Anderson's theorem [2] gives

$$d\tilde{X}_\tau = \big(-\tilde{X}_\tau - 2\nabla \log \tilde{\rho}_\tau(\tilde{X}_\tau)\big)\, d\tau + \sqrt{2}\, d\bar{W}_\tau,$$

run from $\tau \to \infty$ back to $\tau = 0$. Now apply the time change $t = e^{-\tau}$, so $\tau = 0$ corresponds to $t = 1$ (data) and $\tau \to \infty$ to $t = 0$ (noise), matching the SI convention. Since $d\tau = -dt/t$ and $\sqrt{2}\, d\bar{W}_\tau = \sqrt{2/t}\, d\bar{W}_t$, the reverse SDE becomes, in the t variable,

$$dX_t = \frac{1}{t}\big(X_t + 2\nabla \log \rho_t(X_t)\big)\, dt + \sqrt{\frac{2}{t}}\, d\bar{W}_t,$$

which is exactly the SI forward SDE (35) with diffusivity $\epsilon_t = 1/t$. To verify, choose $(\alpha_t, \beta_t) = (\sqrt{1 - t^2},\, t)$, so that $\alpha_t \gamma_t = 1$ and the score formula (31) gives $\nabla \log \rho_t(x) = t\, b_t(x) - x$. Substituting into (34) with $\epsilon_t = 1/t$:

$$b^{1/t}_t(x) = \left(1 + \frac{1}{t} \cdot t\right) b_t(x) - \frac{1}{t}\, x = 2b_t(x) - \frac{x}{t},$$

which matches $\frac{1}{t}(x + 2\nabla \log \rho_t(x)) = \frac{1}{t}(x + 2(t b_t(x) - x)) = 2b_t(x) - x/t$ exactly.

Notably, $\epsilon_t = 1/t$ is precisely the optimal diffusivity for this interpolant in the sense of Proposition 3.4. The singularity at $t = 0$ is simply the image of $\tau \to \infty$ under the compactification $t = e^{-\tau}$: the OU process runs for infinite pseudo-time, and the time change compresses this into the finite interval $[0, 1]$, necessarily producing a singularity at the noise endpoint. The SI framework makes clear that this singularity is an artifact of the time-reversal procedure and not intrinsic to the generative task: since the path design and the diffusivity are decoupled, one may freely replace $\epsilon_t = 1/t$ by any non-singular $\epsilon_t > 0$ and still recover the correct marginals ρ_t, at the cost of no longer sampling with the optimal diffusivity.

4. Generalizations

4.1. Conditional Generation

In many applications the data comes in pairs (x_1, ξ) drawn from a joint distribution $\mu_1(dx_1, d\xi)$, and the goal is to sample the conditional distribution $\mu_1^c(dx_1|\xi)$ for a given ξ (a class label, a corrupted observation, a low-resolution image, etc.). The SI framework extends to this setting with no structural change: one builds the interpolant $I_t = \alpha_t x_0 + \beta_t x_1$ exactly as before, and the continuity equation (4) still holds for the conditional density $\rho_t(\cdot|\xi)$. The only modification is that the drift and the denoisers become functions of ξ:

$$b_t(x, \xi) = \mathbb{E}[\dot{I}_t \mid I_t = x,\ \xi], \qquad \eta_t^{0,1}(x, \xi) = \mathbb{E}[x_{0,1} \mid I_t = x,\ \xi],$$

and the quadratic regression objectives (**??**) and (**??**) are minimized over networks that take ξ as an additional input. At inference time, one fixes ξ and integrates the ODE or SDE (39) with $b_t(\cdot, \xi)$ to draw a sample from $\mu_1(\cdot|\xi)$. The same ϵ_t-family of stochastic samplers and the optimal diffusivity of Proposition 3.4 carry over unchanged.

4.2. Operator-Valued Interpolants and Multiscale Modeling

A natural extension of the scalar interpolant $I_t = \alpha_t x_0 + \beta_t x_1$ is to allow α_t and β_t to be linear operators acting on $\mathbb{R}^d$. For instance, one may take $\alpha_t = \mathrm{diag}(\alpha_t^{(1)}, \dots, \alpha_t^{(d)})$ and $\beta_t = \mathrm{diag}(\beta_t^{(1)}, \dots, \beta_t^{(d)})$, where each coordinate k follows its own schedule $(\alpha_t^{(k)}, \beta_t^{(k)})$. This allows different components of x to be interpolated on different timescales, which is natural in multiscale problems where coarse-scale features should be generated before fine-scale ones. The continuity equation and the quadratic regression objectives for b_t remain formally identical, with $\dot{\alpha}_t, \dot{\beta}_t$ now interpreted as operator derivatives. The theory of Section **??** extends to this setting [**?**].

4.3. Multitask Interpolants

The most general version of this idea replaces the time parameter $t \in [0,1]$ by a pair of linear operators (α, β) drawn from some distribution π over a set $\mathcal{A}$ of admissible operator pairs [**?**]. The generalized interpolant is

$$I_{\alpha,\beta} = \alpha x_0 + \beta x_1,$$

with density $\rho_{\alpha,\beta}$, where (α, β) is drawn from a distribution π over a set $\mathcal{A}$ of admissible operator pairs [**?**]. The set $\mathcal{A}$ should be such that: (i) it contains pairs with $\alpha = 0$ (pure x_1) and $\beta = 0$ (pure x_0), so that both endpoints are represented in training; and (ii) $\alpha + \beta$ is never identically zero, ensuring that $I_{\alpha,\beta}$ is always informative. One then seeks to learn the conditional expectations

$$\eta^0_{\alpha,\beta}(x) = \mathbb{E}[x_0 \mid I_{\alpha,\beta} = x,\ \alpha, \beta], \qquad \eta^1_{\alpha,\beta}(x) = \mathbb{E}[x_1 \mid I_{\alpha,\beta} = x,\ \alpha, \beta].$$

These can be learned separately by minimizing two quadratic objectives:

$$\mathcal{L}^k(\eta^k) = \mathbb{E}_{(\alpha,\beta)\sim\pi}\, \mathbb{E}_{x_0,x_1}\left[|\eta^k_{\alpha,\beta}(I_{\alpha,\beta}) - x_k|^2\right], \qquad k = 0, 1.$$

Note that η^0 and η^1 are not independent: since $\mathbb{E}[I_{\alpha,\beta} \mid I_{\alpha,\beta} = x,\ \alpha, \beta] = x$ by definition, we have the identity

$$\alpha\, \eta^0_{\alpha,\beta}(x) + \beta\, \eta^1_{\alpha,\beta}(x) = x,$$

so that either denoiser determines the other, and only one regression needs to be performed in practice.

When $x_0 \sim \mathcal{N}(0, I_d)$ independently of x_1, the score of $\rho_{\alpha,\beta}$ can be expressed directly in terms of η^0 via Stein's formula:

$$\nabla \log \rho_{\alpha,\beta}(x) = -\alpha^{-1} \eta^0_{\alpha,\beta}(x).$$

To perform generation, one chooses a time-dependent path $t \mapsto (\alpha_t, \beta_t)$ in $\mathcal{A}$ with (α_0, β_0) corresponding to pure noise and (α_1, β_1) to pure data, and integrates the probability flow ODE obtained by differentiating I_{α_t,β_t} along the path. By the chain rule, the drift is

$$b_t(x) = \dot{\alpha}_t\, \eta^0_{\alpha_t,\beta_t}(x) + \dot{\beta}_t\, \eta^1_{\alpha_t,\beta_t}(x),$$

which is the direct analogue of (**??**). When $x_0 \sim \mathcal{N}(0, I_d)$, the score formula above gives access to the full family of stochastic samplers (39) without additional training. The key advantage of the multitask framework is that once $\eta^{0,1}$ are learned over π, one can perform *zero-shot* inference for any operator path $(\alpha_t, \beta_t)_{t\in[0,1]}$ supported on $\mathcal{A}$, enabling conditional generation, inpainting, posterior sampling, and multiscale modeling from a single trained model.

A. Notation and Conventions

Throughout these notes, we work in $\mathbb{R}^d$. We denote the *unknown target distribution* by $\mu_1(dx)$ and the *base distribution* by $\mu_0(dx)$. When these distributions are absolutely continuous with respect to the Lebesgue measure, we write their densities as $\rho_1(x)$ and $\rho_0(x)$, respectively, so that

$$\mu_k(dx) = \rho_k(x)dx, \qquad k = 0, 1.$$

More generally, we will work with *time-dependent measures* $\mu_t(dx)$, parametrized by $t \in [0, 1]$, with density $\rho_t(x)$ whenever it exists:

$$\mu_t(dx) = \rho_t(x)dx, \qquad t \in [0, 1].$$

The boundary conditions $\mu_{t=0} = \mu_0$ and $\mu_{t=1} = \mu_1$ encode the interpolation problem: we seek a dynamical process that continuously deforms the base into the target.

Notation conventions. Throughout these notes we maintain a strict distinction between two kinds of objects.

- *Fixed points* $x, x_0, x_1 \in \mathbb{R}^d$ denote deterministic elements of $\mathbb{R}^d$, typically obtained as samples from a distribution or as the argument of a function. Lower-case letters are used for these.
- *Stochastic processes and ODE solutions* X_t denote random variables or trajectories evolving in time. Upper-case letters are used for these. When X_t solves the ODE $\dot{X}_t = b_t(X_t)$, the subscript $t = 0$ or $t = 1$ specifies which endpoint is prescribed: we write $X_{t=0} = x_0$ if the *initial* condition is fixed (the ODE is integrated forward from x_0), and $X_{t=1} = x_1$ if the *final* condition is fixed (the ODE is integrated backward from x_1). The notation $X_t(x_0)$ or $X_t(x_1)$ makes the dependence on the prescribed endpoint explicit when needed.

We will test measures against smooth, bounded functions. Specifically, we say $\phi : \mathbb{R}^d \to \mathbb{R}$ is a *test function* if it is bounded and twice continuously differentiable; we write $\phi \in C_b^2(\mathbb{R}^d)$, where

$$C_b^2(\mathbb{R}^d) = \left\{ \phi \in C^2(\mathbb{R}^d) \,:\, \phi, \nabla\phi, \nabla^2\phi \text{ are bounded} \right\}.$$

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